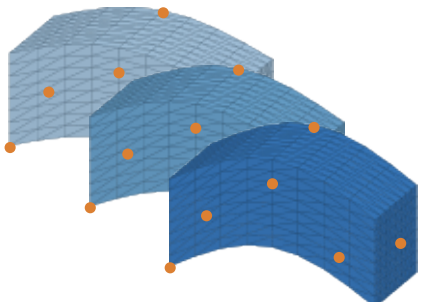
## 01 OBSERVE



**A short position-track prefix**

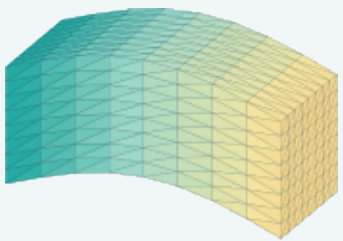
Known geometry + mass

Fit material / attachment descriptors

## 02 COMPOSE LOCAL MECHANICS

### Material that moves

$z_m(X) \longrightarrow W_\theta(F, z_m)$



Spatial material field

### Frozen decoders



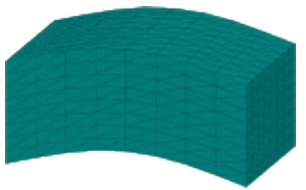
Energy → stress

Q2 mechanics → motion

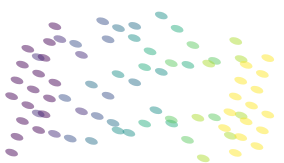
**INTERACTIONS** Tool poses • attachments • contact • gravity

$$\Pi = \int W_\theta \, dV + \int \psi_\phi \, dA + \Pi_{contact} - f_g^\top u$$

## 03 PREDICT



**New deformation + reactions**



**Transport Gaussian appearance**

Scene fitting changes descriptors, not shared network weights. Illustrations are schematic; physical and image errors are evaluated separately.